

An LLM-Agent and Knowledge Graph–Integrated Framework for Collaborative Optimization of Hydropower Maintenance Resources Scheduling

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Abstract—Efficient scheduling of maintenance resources for hydropower units is crucial for improving maintenance efficiency and reducing costs. However, traditional methods often rely on manual experience and static scheduling rules, lacking dynamic optimization mechanisms. To address these limitations, this paper proposes an LLM-Agent and knowledge graph-integrated framework for collaborative optimization of hydropower maintenance resources scheduling. First, an LLM-assisted approach is employed to construct a maintenance knowledge graph. Then, a query-aware subgraph retrieval method is designed to extract relevant triplets by identifying core entities in user queries and leveraging semantic alignment strategies for multi-hop subgraph retrieval. Subsequently, a three-stage resource allocation generation mechanism based on prompt engineering is proposed. This mechanism utilizes the retrieved knowledge subgraphs to guide the generation of diverse and structured maintenance plans. Finally, an LLM-Agent-based optimization module, incorporating a genetic algorithm, is developed to dynamically manage resource constraints and reduce costs across multiple planning alternatives. Experimental results demonstrate that the proposed framework significantly enhances the efficiency of maintenance resource allocation, offering a novel paradigm for advancing the intelligent hydropower equipment maintenance.

Keywords—Hydropower Maintenance Resource Scheduling; Large Language Model; Knowledge Graph; Agent Optimization

I. INTRODUCTION

With the increasing complexity of equipment in hydropower stations, the rational scheduling and allocation of maintenance resources has become a key factor in ensuring the safe and stable operation of power systems. Traditional maintenance resource planning methods rely heavily on manual expertise and static scheduling rules, making them insufficient to cope with dynamic operational environments and evolving resource constraints. As a result, it often leads to resource mismatches and maintenance delays [1].

Recently, large language models (LLMs) have demonstrated strong capabilities in language understanding and logical reasoning, offering a feasible approach for intelligent resource scheduling [2]. Meanwhile, knowledge

graph (KG), as a structured form of knowledge representation, provides LLMs with rich external knowledge, thereby enhancing their reasoning and generation abilities [3]. In the field of equipment maintenance, KGs are widely used in scenarios, such as fault diagnosis and maintenance strategy recommendation, significantly improving knowledge management and reasoning performance [4]. For example, Wei et al. employed a retrieval-augmented generation (RAG) strategy by feeding entities and relations from KGs as textual prompts into LLMs to improve reasoning accuracy and reliability [5]. Yu et al. aligned LLMs with KG through dual-view knowledge graph contrastive learning, thereby enhancing the performance of KG completion tasks [6]. However, most existing approaches focus on static knowledge alignment, lacking dynamic retrieval capabilities that adapt to specific user queries or tasks.

At the application level of knowledge graph, Gijre et al. proposed a novel technique to improve LLMs for document analysis. By employing a retrieval-augmented generation strategy, they enhanced the accuracy and contextual understanding of question-answering systems [7]. Cai et al. proposed the Similar Graph Enhanced Retrieval-Augmented Generation method, which transforms user queries into graph patterns using an LLM and matches them with candidate subgraphs based on graph semantic distance measures, thereby enabling efficient subgraph retrieval [8]. Tao et al. introduced the LLM framework, which integrates hierarchical task agent with instruction-level retrieval-augmented generation techniques to enhance the flexibility and accuracy of LLMs in handling both known and novel maintenance objects and planning scenarios [9]. Zhang et al. presented a RAG framework that combines LLMs with KG. In the context of new energy vehicles and hydropower station maintenance, the framework integrates KG and LLMs to achieve intelligent fault diagnosis and resource scheduling [10]. While these methods improve retrieval and reasoning performance, they often rely on pre-defined templates or static entity matching, which may limit their adaptability to complex maintenance scenarios requiring deeper semantic understanding and task-specific contextualization.

In addition, research on building autonomous agent based on LLMs has been gaining momentum. This agent aims to leverage the extensive knowledge and reasoning capabilities of LLMs to achieve human-like decision-making and problem-solving abilities [11]. Compared to traditional reinforcement learning agent, LLM-based agent offers notable advantages in knowledge representation completeness, natural language interaction, and system interpretability. For instance, Schick et al. proposed the Toolformer model, integrating self-supervised learning with API-calling techniques to enhance tool usage capabilities of language models across a variety of downstream tasks [12]. Similarly, Sakhinana et al. developed a Process Engineering Operations Assistant (PEOA) framework, a modular expert system coordinated by a meta-agent. By employing high-level knowledge modeling and graph retrieval, PEOA enables task decomposition and dynamic problem-solving [13]. Although valuable research has been conducted on LLM-based agent, its practical application to hydropower unit maintenance resource optimization remains limited. Thus, effective strategies are further explored and developed to integrate optimization methods into LLMs.

To address the above challenges, this paper proposes an LLM-Agent and knowledge graph-integrated framework for collaborative optimization of hydropower maintenance resources scheduling. The framework enables a closed-loop process for maintenance knowledge modeling, resource allocation generation, and decision optimization. The main contributions of this study are as follows:

- 1) A query-aware subgraph retrieval method is designed. It uses an LLM to identify key entities in user queries and retrieves relevant subgraphs from the knowledge graph through a multi-hop semantic alignment strategy. This approach helps to extract structured contextual information closely related to the current task, enhancing the relevance and usability of maintenance knowledge.
- 2) A three-stage prompt-based mechanism for generating resource allocation plans is proposed. Conditioned on the retrieved knowledge subgraphs, the rule-based matching is firstly applied to extract triples containing key fields. By combining the multi-step prompt engineering strategy, the LLM is guided to generate various structured maintenance resource allocation plans. Finally, the filtered triples from the first step and the generated plans are used as context inputs to the LLM, evaluating their confidence levels. It enables verification of the factuality and structural consistency of the generated plans.
- 3) An integrated decision-making framework is developed, combining evolutionary algorithms with LLM-Agent. By utilizing the LLM's tool-use capabilities, the framework designs callable dynamic resource constraints and genetic algorithm operators. It enables dynamic cost optimization across multiple maintenance plans. The final output is a resource allocation solution that balances feasibility and economic efficiency, generated with the aid of the LLM.
- 4) A resource optimization decision-making framework integrating a genetic algorithm and LLM-Agent is developed. By leveraging the function-calling capability of LLM, tool parameters were designed for invocation by the LLM, allowing dynamic resource constraints to be passed into the genetic algorithm. This enables cost optimization of multiple maintenance resource plans

within the topological maintenance workflow. Finally, the LLM combines the optimization results to generate resource allocation schemes that balance feasibility and cost-effectiveness.

The remainder of this paper is organized as follows. Section 2 describes the proposed framework and its specific methods. Section 3 presents the experimental validation. Section 4 is the conclusion.

II. METHODOLOGY

The process of the proposed knowledge graph and LLM-Agent collaborative optimization framework is shown in Fig. 1. The following subsections detail its technical aspects.

A. KG Construction and Query-Guided Subgraph Retrieval Based on LLM

This section focuses on constructing a maintenance KG from original maintenance procedure data. LLM-assisted entity recognition and graph alignment methods are used to complete semantic parsing. A multi-level subgraph retrieval strategy is implemented within the maintenance KG.

1) Maintenance Knowledge Graph Construction

A locally-deployed LLM assists in extracting key information from maintenance procedure PDF files into structured data. The extracted data is stored in CSV format and imported into the Neo4j graph database via a Python script, completing the construction of the unit maintenance knowledge graph. The structure of the KG for unit maintenance is shown in Fig. 2.

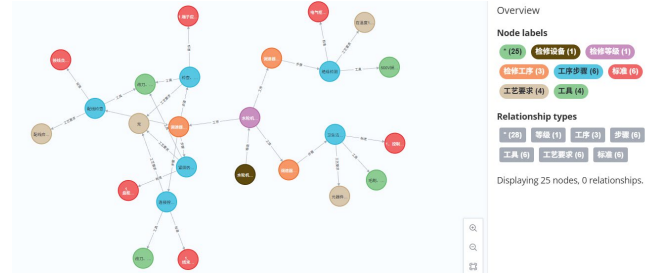


Figure. 2. Display of the KG for Unit Maintenance

2) Entity Recognition and Alignment

LLM is used to identify key maintenance entities from query texts. The identified entities and nodes in the graph are vectorized using a semantic vector model. Cosine similarity of the vector matrices is calculated to determine the most relevant nodes in the graph.

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \cdot \sqrt{\sum_{i=1}^n B_i^2}} \quad (1)$$

where A and B represent the n -dimensional feature vectors of two texts, respectively.

3) Subgraph Retrieval and Generation

The Top-K nodes obtained through vector similarity calculation are used to extract subgraphs of up to three hops based on node relevance and graph structure. Finally, maintenance solution-related triple information is output.

$$G_{sub} = \bigcup_{v \in V_{top}} BFS(v, depth = 3) \quad (2)$$

where BFS represents breadth-first search, with $depth=3$ limiting the search to three hops.

B. LLM-Driven Generation of Maintenance Plans Based on Knowledge Subgraphs

1) Rule Matching and Screening

The retrieved triple data is extensive and may include some irrelevant triples, introducing noise that could distract the attention mechanism of the LLM. Therefore, before inputting the triple data into the LLM, rule-based matching and filtering are applied to select triples containing specific information.

$$T_{\text{filtered}} = \left\{ (h_i, r_i, t_i) \in T \mid \prod_{j=1}^k \phi(h_i, r_i, t_i) = 1 \right\} \quad (3)$$

where $T = \{(h_i, r_i, t_i) \mid i = 1, 2, \dots, N\}$ represents the original triples, $\phi: (h, r, t) \rightarrow \{0, 1\}$ denotes the rule-matching function.

2) Maintenance Resource Plan Generation by Prompt Engineering

The filtered triples are used as the input context for the LLM. Through instruction prompts and one-shot prompt design, the LLM is guided to output structured maintenance plans.

3) Self-Verification by Dual LLMs

Considering the randomness and hallucination issues in LLM outputs, the authenticity of the generated maintenance plans is ensured by comparing the model-generated plans with atomic graph triples as context inputs to the LLM. The LLM evaluates the confidence level of the generated plans, enabling self-verification.

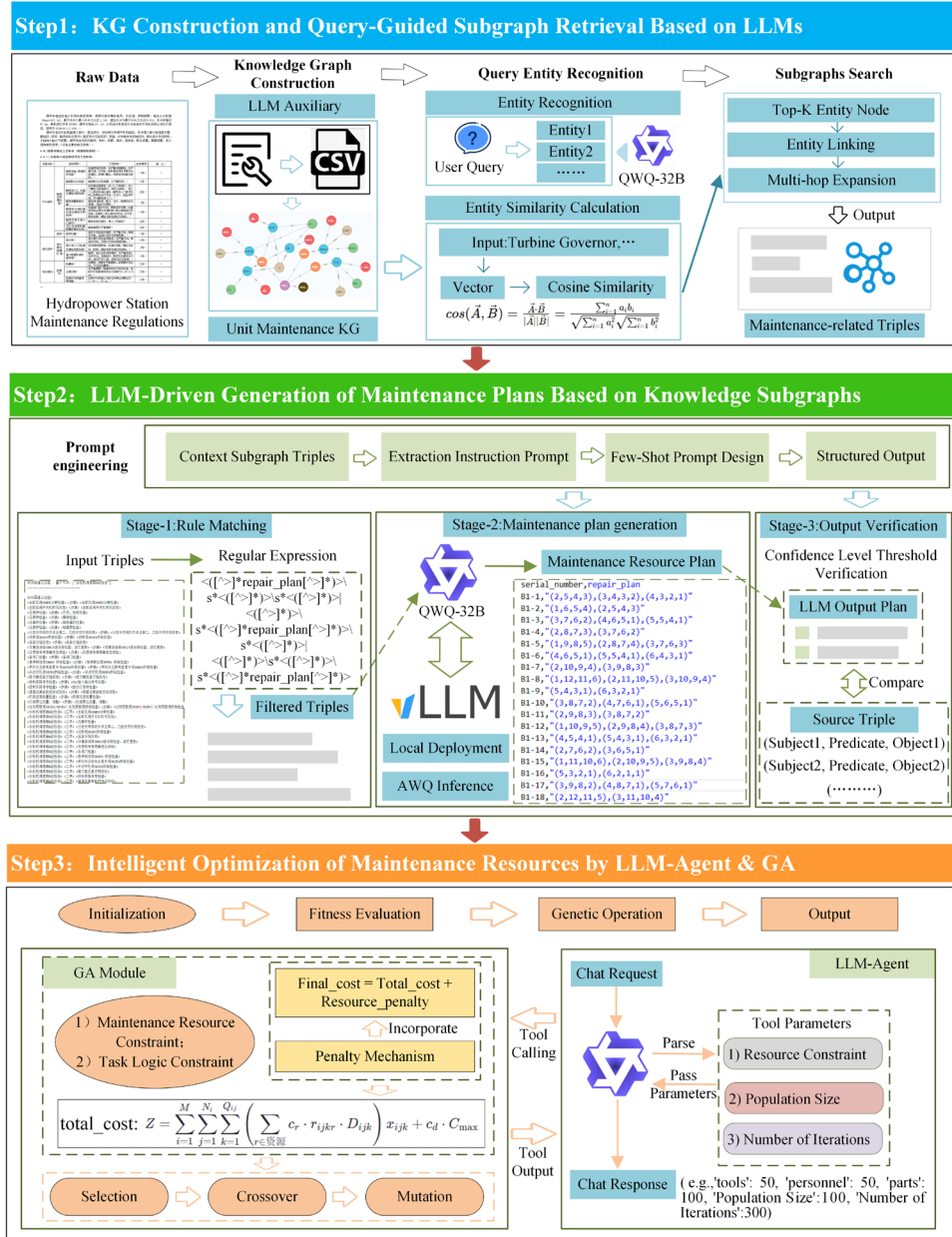


Figure. 1. Flowchart of the proposed framework

C. Intelligent Optimization of Maintenance Resources by LLM-Agent & Genetic Algorithm

1) Problem Description and Modeling

a) Hydropower unit equipment maintenance is divided into multiple levels (e.g., Level B maintenance). Each level consists of M maintenance processes. The i -th process includes N_i maintenance steps, and the j -th step corresponds to Q_{ij} optional maintenance plans. Precedence constraints exist between different steps, meaning some steps can only begin after the preceding steps are completed.

b) Maintenance resource types include tools, personnel, and spare parts. Each maintenance plan corresponds to a unique combination of resources. These resources are renewable and can be reused after consumption. Instantaneous resource usage at any time cannot exceed the available resource constraints.

c) The optimization objective function for maintenance resources includes maintenance costs and labor hours. To simplify calculations, a conversion coefficient c_d is introduced to transform the multi-objective model into a single-objective model by converting labor hours into loss costs caused by maintenance downtime. It unifies the dimensions of the objective function and resolves convergence issues in multi-objective models. The final optimization objective function is as follows:

$$\text{Minimize } Z = \sum_{i=1}^M \sum_{j=1}^{N_i} \sum_{k=1}^{Q_{ij}} \left(\sum_{r \in \text{resource}} c_r \cdot r_{ijk} \cdot D_{ijk} \right) x_{ijk} + c_d \cdot C_{\max} \quad (4)$$

where Z represents the total maintenance cost, c_r is the unit usage cost of resource r , r_{ijk} denotes the usage of resource r by plan k in step j of process i , D_{ijk} represents the duration of plan k in step j of process i , and $x_{ijk} \in \{0,1\}$ indicates whether plan k for step j of process i is selected. c_d is the downtime loss cost, and $C_{\max}$ is the total maintenance labor hours.

2) Genetic Algorithm Module Design

To optimize the maintenance resource objective function, a genetic algorithm-based solution method is designed and implemented. It minimizes the total maintenance cost while satisfying task dependencies and resource constraints.

a) Each chromosome represents a complete set of maintenance step plans, with its length equal to the total number of steps. Each gene in the chromosome indicates the selected maintenance plan index for a specific step. The initial population is generated by randomly selecting plans for each task, ensuring good diversity within the population.

b) The fitness function evaluates individuals based on the total maintenance cost. Considering the dynamic constraints of available resources, a resource constraint mechanism is incorporated into the fitness calculation. If an individual violates resource limits, a large penalty value is added to its total cost, making it inferior to feasible solutions during selection. It guides the search to converge toward the optimal solution that satisfies the constraints. The fitness function is defined as follows:

$$f(\mathbf{x}^{(i)}) = \frac{1}{Z(\mathbf{x}^{(i)}) + P(\mathbf{x}^{(i)}) + \varepsilon} \quad (5)$$

where $Z(\mathbf{x}^{(i)})$ represents the total cost of the plan encoded by the chromosome, $P(\mathbf{x}^{(i)})$ denotes the penalty value λ if resource constraints are violated, otherwise it is 0.

c) To ensure the feasibility of the maintenance process, task dependencies are explicitly modeled in the algorithm. When evaluating individuals, the labor hours required for

each maintenance step in the plan are parsed, and a completion time calculation logic based on a dependency graph is constructed. The earliest completion time for each task is calculated through topological traversal, determining the total labor hours.

d) The roulette wheel selection is used for parent selection, ensuring higher-fitness individuals have a greater chance of reproduction. A single-point crossover strategy is applied with a probability of 0.8 to exchange chromosome segments at random positions, generating two offspring. Mutation operates with a probability of 0.1 at each locus to randomly change the selected plan for a task, avoiding premature convergence to local optima.

e) The algorithm iteratively executes parameters parsed and invoked by the LLM tool, retaining the current best solution in each generation and generating a new population. If no better feasible solution is found after several consecutive generations, the search is considered to have converged. The optimal maintenance plan that satisfies resource constraints is output ultimately.

3) Design of LLM-Agent Based on Tool Call

This section constructs an agent that integrates LLM with a genetic algorithm through tool invocation. The agent aims to assist in the optimal allocation and scheduling of unit maintenance resources.

a) The tool function interface is described using JSON schema. It strictly defines the types and structures of input parameters to ensure the LLM can accurately generate invocation statements. The function description is passed to the LLM through the tools field to guide it in generating correct invocation requests.

b) After the user sends a request, the agent sends the input along with the historical dialogue context to the LLM. If the model determines that a tool function needs to be invoked, it returns a structured response (tool_calls field) containing the function invocation intent. The agent parses this invocation, executes the corresponding backend algorithm for optimization, and encapsulates the optimization results into a tool response message fed back to the LLM. This achieves comprehensive result response output.

III. RESULTS

A. Data Source and KG Construction

This paper selects the maintenance regulations of a certain hydropower plant in the Dadu River Basin. The content covers the maintenance regulations for the mechanical and electrical parts of equipment such as the water turbine, speed governing system, and generator. Some maintenance resource data are based on reasonable assumptions. Taking the maintenance of the speed governing system as an example, a maintenance knowledge graph is constructed, and the main parameters of the graph are shown in Table I.

TABLE I MAIN PARAMETERS OF THE KG

Parameter Name	Values
Number of nodes	217
Node label	7
Number of relationships	389
Relationship type	6

B. Results of Subgraph Retrieval and Generation

1) LLM Deployment and Configuration

This paper adopts the vLLM service library to locally deploy the AWQ-quantized version of QWQ-32B. The deployment is achieved using 2 NVIDIA GeForce RTX 4090D GPUs.

2) Results of Retrieval and Generation

For the query text, "Under the constraints of tools, personnel, and spare parts being 50, 50, and 100 respectively, please provide the optimal resource allocation plan for Class B maintenance of the governor," the LLM identifies the maintenance entities "governor" and "Class B maintenance." Subsequently, subgraph retrieval, maintenance plan generation, and self-verification are performed. Finally, 248 relevant triples, 38 maintenance steps, and $3^{16} \times 2^{22}$ maintenance plans are obtained.

C. Experiment on Maintenance Resource Optimization

Based on the LLM-Agent maintenance resource optimization method integrated with a genetic algorithm described in Section 2.3, and using the above query text as an example, the genetic algorithm parameters adopt the default configuration if not specified. The analytical results and the optimization process are shown in Table II and Figure 3.

TABLE II PARAMETERS OF THE ANALYTICAL TOOL

Parameter Name	Values
Resource Constraint	Tools 50
	Personnel 50
	Parts 100
GA Population Size	100
GA Number of Iterations	300



Figure 3. LLM-Agent Optimization Process and Results Presentation

On the left side of the Figure 2, the user provides specific resource constraints and optimization problems (blue dashed box). The agent automatically parses the tool parameters from the query and passes them to the algorithm for iterative optimization (orange box). Finally, the optimized results are sent back to the LLM for comprehensive output (green box).

The changes in optimal cost and Pareto front during the algorithm iteration are shown in Figures 4 and 5. Experimental results show that the LLM-Agent integrated with a genetic algorithm achieves full-process automation from natural language queries to cost evaluation and selection of multiple plans. By incorporating dynamic maintenance resource

constraints, it generates resource allocation suggestions that balance feasibility and economic efficiency, effectively improving the efficiency of maintenance resource allocation.

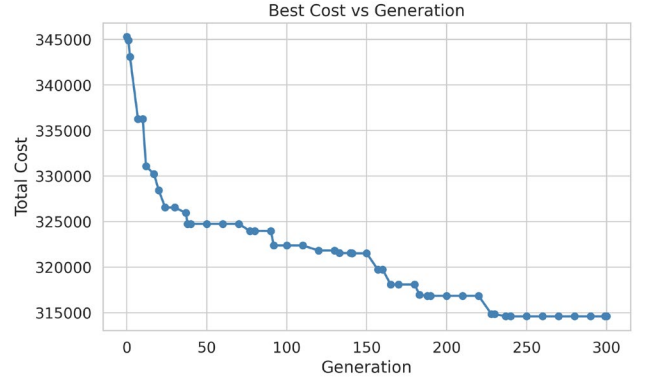


Figure 4. Cost Iteration Process

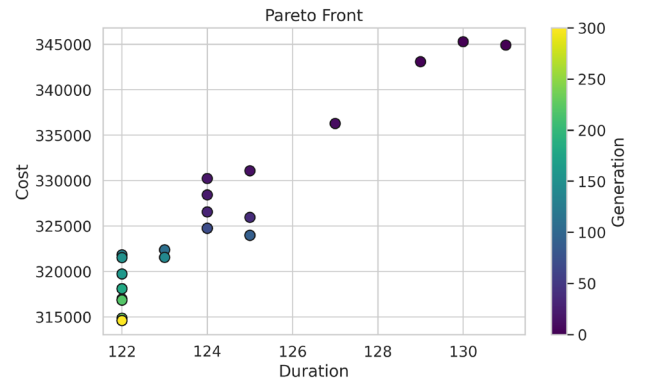


Figure 5. Pareto Front Diagram

IV. CONCLUSION

This paper investigates key issues in intelligent scheduling of resource allocation for hydropower unit maintenance. It addresses challenges such as reliance on manual experience, lack of flexibility in static scheduling rules, and variable constraints on available maintenance resources. A collaborative optimization framework integrating KG and LLM-Agent is proposed. This framework includes multiple stages: KG construction, subgraph retrieval, maintenance plan generation, and optimization decision-making. It establishes a full-process collaboration from knowledge modeling to plan generation and optimization decisions. The query-aware subgraph retrieval mechanism and the multi-stage resource generation method based on prompt engineering enhance the usability of knowledge graph and improve the relevance and accuracy of content generated by LLM. By combining the function-calling capability of LLM with the optimization power of evolutionary algorithms, comprehensive cost optimization of maintenance plans is achieved. This approach significantly enhances the intelligence level of resource scheduling for hydropower unit maintenance. Experimental validation shows that the framework is practical and scalable, providing an expandable technical pathway for intelligent scheduling in complex industrial maintenance scenarios.

Future research will focus on incorporating multimodal data from hydropower stations (e.g., time-series data and images), expanding the coverage of the maintenance

knowledge graph for hydropower units, and developing multifunctional maintenance agents to enable more advanced intelligent maintenance and autonomous decision-making capabilities.

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